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2 August 1965
SUPERSEDING 1/
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1 March 1965

MILITARY SPECIFICATION

GARBAGE DISPOSAL MACHINES, COMMERCIAL, VERTICAL TYPE, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope.- This specification covers vertical type commercial garbage disposal machines, for automatic grinding and disposal of food scraps aboard Naval ships.

1.2 Classification.- Garbage disposal machines shall be of the following sizes as specified (see 6.1):

SIZES
I - 400 pounds per hour.
II - 1000 pounds per hour.
III - 2000 pounds per hour.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 10016, N.Y.)

AMERICAN STANDARDS ASSOCIATION (ASA)
C1 - National Electrical Code.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 10016, N.Y.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York 10017, N.Y.)

3. REQUIREMENTS

3.1 Preproduction samples.- Prior to beginning production a sample model disposal machine of each size shall be tested as specified in 4.3 (see 6.2).

3.2 Material.- Material used shall be equal to or better than the quality used by the manufacturer in his regular commercial production. Materials used in sizes I and II machines shall be suitable for use with fresh water. Materials used in the size III machine shall be suitable for use with fresh water and sea (salt) water.

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 4540

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3.3 Design. - The design of the machines shall be in accordance with the best engineering practice. Sizes I and II machines shall be designed to grind garbage and wash it down the drain when using not more than 3 gallons of water per minute through the grinder body (additional water may be used below the grinder to keep the drain clear). The size III machine shall have the same performance as sizes I and II except that not more than 5 gallons of water per minute shall be used. Ground garbage shall not pack or adhere to any parts of the machine when water is flowing. Garbage shall be ground to a finely divided mash or pulp, 40 percent of which shall pass through a number 8 sieve and not less than 60 percent through a number 3-1/2 sieve. Sieves shall be U.S. Standard sieve series and percentages shall be based on drained weights. Machines shall not jam when operated at full capacity.

3.3.1 Cutting Mechanism. - The cutting mechanism shall be composed of the impeller (rotor) and the cutting ring. The impeller and all rotating parts shall be accurately balanced to reduce vibration and insure quiet running machines at designed speeds. The cutting mechanism shall be of alloy steel. Carburizing, cemented carbides and weld-on hard facing alloys may be used. The hardness of the cutting mechanism shall be not less than 50 nor more than 62 on the Rockwell "C" scale. Cemented carbides, when used, shall be as resistant to the attack of food acids and alkalies as the alloy steel used.

3.4 Construction. - Each machine shall consist essentially of a grinding chamber containing the cutting edges, an impeller or rotor, and the electric driving motor. Assembly shall be such, either by separate motor enclosure or shield, that water cannot enter the motor in event of grinder seal failure. The grinding chamber shall be designed to prevent garbage or water being thrown back out of the machine. A silverware trap shall not be provided. Size I machines shall be suitable for mounting to a sink with sink adaptor, or conical hopper (size of drain opening shall be specified (see 6.1)). Unless otherwise specified in the contract or order, conical hopper shall not be supplied with the type I machine. Sizes II and III machines shall be provided with means of supporting the units from the deck (to be free-standing).

3.4.1 Rotation direction. - The designed direction of rotation shall be clearly indicated by means of a circular arrow cast or permanently attached on the grinder housing. Machines which may be operated in either direction need not be so marked.

3.4.2 Shaft seal. - The grinder shaft shall be fitted with a positive seal to prevent water leaking from the grinding chamber along the shaft. The seal shall be protected against the entry of grit or dirt.

3.5 Bearings. - All bearings for rotating elements shall be permanently lubricated anti-friction type.

3.6 Discharge. - The discharge outlet of each machine shall be such that it can be connected to standard pipe of not less than 1-1/2-inch size.

3.7 Flow interlock. - A flow interlock shall be provided with each machine for installation by others.

3.8 Garbage feed chute. - Unless otherwise specified in the contract or order, all machines shall be furnished without garbage feed chute or hopper.

3.9 Electrical requirements. - Electrical equipment shall be designed for operation on an ungrounded electrical system but shall operate satisfactorily and without hazard to personnel or equipment with an accidental ground on any conductor.

3.9.1 Motors. - Motors shall conform to NEMA Publication MG1, and shall have the following characteristics:

Ambient temperature	40°C.
Enclosure	Dripproof.
Duty	Continuous.
Horsepower	As required for satisfactory performance.
Electrical characteristics	As specified (see 6.1).
Bearings	Ball or roller.
Insulation	Class A or B.

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3.9.2 Motor control.- Motors shall be provided with either built-in motor overload protection for single phase and with magnetic motor starters with heaters for 3 phase.

3.9.3 Starting controls and wiring.- Starting controls and wiring shall conform to the requirements of ASA Publication C1.

3.10 Identification Plate.- Each machine shall have an identification plate located in a readily accessible location, on either the motor or the machine, giving the model and serial number of the machine, the type of current, voltage, phase and frequency (if alternating current) and rated horsepower. Identification plate shall be made of noncorrosive metal.

3.11 Instruction plate.- A noncorrosive metal plate, for installation by others, shall be provided. This plate shall contain instructions describing any special or important procedure to be followed in operating or servicing the machine.

3.12 Finish.- The equipment shall be finished and painted in accordance with the manufacturer's standard practice.

3.13 Manuals.- Manuals shall be the standard commercial manufacturer's instructions including manufacturer's illustrations or drawings to concisely outline the proper installation, operation, maintenance and parts identification of the garbage disposal machine furnished. The manuals shall be suitably bound in a self-cover of durable quality. Two copies of the manual shall be packed with each machine. Two copies of the manual shall be submitted to the Bureau of Ships, via the Inspector of Naval Material, for assignment of a NAVSHIPS number. Additional copies for stock shall be as specified (see 6.1).

3.14 Repair parts and maintenance tools.- Onboard repair parts and maintenance tools shall be as specified in the contract or order as selected from the list recommended by the manufacturer to maintain operation of each unit for one year.

3.15 Workmanship.- Workmanship shall conform to the following:

(a) Welds, where used, shall be cleaned and ground to present a smooth and uniform finish.
(b) All surfaces shall be free of burrs and sharp edges.
(c) Castings shall be smooth and free of blow holes.
(d) Faying surfaces shall be properly machined for uniform fit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample.- Prior to the start of production and within 45 days after the date of award of the contract or order, one machine of each size under contract or order shall be submitted to the U. S. Naval Supply Research and Development Facility, Bayonne, New Jersey, for preproduction testing. The Contracting Officer shall be notified that the machine is ready for preproduction testing. The preproduction sample is defined as a production line model of the product which the manufacturer will subsequently produce in fulfillment of the contract or order. Approval of the preproduction sample by the U.S. Naval Supply Research and Development Facility shall not relieve the supplier of the responsibility of compliance with the requirements of this specification. Any change or deviations from the preproduction sample shall be submitted for the approval of the bureau or agency concerned. The approved preproduction sample will be used as a guide in the inspection of production units, and shall be presented for delivery as the last unit for delivery under the contract or order.

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4.2.1 Prior to the submission of the preproduction sample for testing, the manufacturer shall subject the machine to the test of 4.3.2. Certification containing actual test data shall be furnished showing that the sample submitted for preproduction tests has successfully passed the test of 4.3.2.

4.3 Preproduction inspection. - Preproduction inspection shall consist of the examination specified in 4.4.1 and tests specified in 4.3.1 through 4.3.4.2.

4.3.1 No-load test. - The machine shall be operated for 1/2 hour, without load, at rated voltage to determine that all parts are properly assembled and in working condition. The power input shall be measured. The no-load speed of direct current powered machines shall be measured.

4.3.2 Insulation resistance. - The insulation resistance of motor, controller and connecting cable shall be measured with a 500 volt, d.c., insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.3.3 Full capacity test. - The machine shall be operated at full capacity for a period of 10 minutes to determine that the design and construction is adequate to handle garbage at the specified rate and, in all respects, conforms to the operating requirements of this specification. The machine shall dispose of not less than one sixth of the rated capacity within the 10 minute test period using not more than the specified volume of water through the grinder body. The capacity shall be determined when grinding the test specimen described in 4.3.3.1. Machine shall be operating when specimen is applied.

4.3.3.1 Test specimen. - The test specimen shall consist of the following mixture (percentages based on drained weight):

(a) 15 percent uncooked beef or pork bones (not over 1-1/2 inches in diameter or width).
(b) 15 percent cooked beef or pork bones (not over 1-1/2 inches in diameter or width).
(c) 10 percent cooked poultry bones.
(d) 10 percent grapefruit rinds cut in half and inserted in such a manner that they could float.
(e) 45 percent raw vegetables and trimmings, including lettuce leaves, cabbage leaves and celery.
(f) 5 percent paper and cellophane, including napkins and wax or plastic coated milk cartons (8 ounce size).

4.3.3.2 Fineness test. - The ground garbage resulting from the full capacity test run specified in 4.3.3 shall be screened to determine its fineness conformance to the requirements of 3.3. Ground garbage may be washed through the sieves, but care shall be exercised to refrain from forcing the particles through the perforations with excess pressure.

4.3.4 Pressure test. - This test shall be conducted as follows:

(a) Make discharge opening pressure tight.
(b) Fill grinding chamber to mid depth with cold fresh water.
(c) Make inlet opening pressure tight and apply 1 pound of pressure to the grinding chamber via a fitting in the inlet closure.
(d) Operate the grinder for a period of 10 minutes.

4.3.4.1 Observe that the area below the shaft seal does not become wet due to extreme failure of the shaft seal while operating under pressure. Should water flow through the seal, as opposed to a slow drip, the machine shall not be offered for delivery.

4.3.4.2 After pressure test, disassemble the motor to the extent necessary to determine whether water has entered the motor enclosure. Water within the motor enclosure shall be reason for rejection.

4.4 Quality conformance inspection. -

4.4.1 Examination. - All machines under contract or order shall be examined to determine compliance with the non-test requirements of this specification. The examination shall include workmanship, availability of all parts on each unit, and that each unit corresponds to the respective approved preproduction sample.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Garbage disposal machines.-

5.1.1.1 Preservation and packaging.- Preservation and packaging, which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.-

5.1.2.1 Each repair part shall be preserved, individually packaged and marked in such a manner as to insure against deterioration and damage in storage.

5.1.2.2 Onboard repair parts shall be suitably packed to insure protection against physical or mechanical damage during direct shipment from the supply source to the using activity. The shipping container or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of machine required (see 1.2).
(c) Size of drain opening for size I machine (see 3.4).
(d) Electrical characteristics (see 3.9.1).
(e) Quantity of manuals required for stock (see 3.13).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 4540-N016SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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